Chapter 18, Part B
Forecasting

- Trend Projection
- Seasonality and Trend
- Time Series Decomposition

Trend Projection

- If a time series exhibits a linear trend, the method of <u>least squares</u> may be used to determine a trend line (projection) for future forecasts.
- Least squares, also used in regression analysis, determines the unique <u>trend line forecast</u> which minimizes the mean square error between the trend line forecasts and the actual observed values for the time series.
- The independent variable is the time period and the dependent variable is the actual observed value in the time series.

Linear Trend Regression

- Using the method of least squares, the formula for the trend projection is:

  - $$T_t = b_0 + b_1 t$$

where: T_t = linear trend forecast in period t

b_0 = intercept of the linear trend line

b_1 = slope of the linear trend line

t = time period

Linear Trend Regression

- For the trend projection equation $T_t = b_0 + b_1 t$

$$b_1 = \frac{\sum_{t=1}^{n}(t-\bar{t})(Y_t-\bar{Y})}{\sum_{t=1}^{n}(t-\bar{t})^2}$$

$$b_0 = \bar{Y} - b_1\bar{t}$$

where: Y_t = value of the time series in period t

n = number of time periods (observations)

$\bar{Y}$ = average values of the time series

$\bar{t}$ = average value of t

Linear Trend Regression

- Example: Auger's Plumbing Service

  - The number of plumbing repair jobs performed by Auger's Plumbing Service in the last nine months is listed on the right.
  - Forecast the number of repair jobs Auger's will perform in December using the least squares method.

Month	Jobs	Month	Jobs
March	353	August	409
April	387	September	399
May	342	October	412
June	374	November	408
July	396		

Linear Trend Regression

(month)	t	$t-\bar{t}$	$(t-\bar{t})^2$	Y_t	$(Y_t-\bar{Y})$	$(t-\bar{t})(Y_t-\bar{Y})$
(Mar.)	1	-4	16	353	-33.67	134.68
(Apr.)	2	-3	9	387	0.33	-0.99
(May)	3	-2	4	342	-44.67	89.34
(June)	4	-1	1	374	-12.67	12.67
(July)	5	0	0	396	9.33	0
(Aug.)	6	1	1	409	22.33	22.33
(Sep.)	7	2	4	399	12.33	24.66
(Oct.)	8	3	9	412	25.33	75.99
(Nov.)	9	4	16	408	21.33	85.32
Sum	45		60	3480		444.00

Linear Trend Regression

- $\bar{t} = 45/9 = 5 \qquad \bar{Y} = 3480/9 = 386.667$
- $$b_1 = \frac{\sum_{t=1}^{n}(t-\bar{t})(Y_t-\bar{Y})}{\sum_{t=1}^{n}(t-\bar{t})^2} = \frac{3480}{60} = 7.12$$
- $b_0 = \bar{Y} - b_1\bar{t} = 386.667 - 7.12(5) = 351.07$
- $T_{10} = 351.07 + (7.12)(10) = 422.27$

Trend Projection

- Example: Auger's Plumbing Service

  - Forecast for December (Month 10) using a three-period (k = 3) weighted moving average with weights of .6, .3, and .1 for the newest to oldest data, respectively. Then, compare this Month 10 weighted moving average forecast with the Month 10 trend projection forecast.

Month	Jobs	Month	Jobs
March	353	August	409
April	387	Septem.	399
May	342	October	412
June	374	Novem.	408
July	396		

Trend Projection

- Three-Month Weighted Moving Average

The forecast for December will be the weighted average of the preceding three months: September, October, and November.

$$F_{10} = .1Y_{Sep.} + .3Y_{Oct.} + .6Y_{Nov.}$$
$$= .1(399) + .3(412) + .6(408)$$
$$= \underline{\underline{408.3}}$$

- Trend Projection

$$F_{10} = \underline{\underline{422.27}} \text{ (from earlier slide)}$$

Trend Projection

- Conclusion

  - Due to the positive trend component in the time series, the trend projection produced a forecast that is more in line with the trend that exists. The weighted moving average, even with heavy (.6) weight placed on the current period, produced a forecast that is lagging behind the changing data.

Holt's Linear Exponential Smoothing

- Charles Holt developed a version of exponential smoothing that can be used to forecast a time series with a linear trend.
- Forecasts for Holt's method are obtained using two smoothing constants, α and β, and three equations.
- Holt's linear exponential smoothing is often called double exponential smoothing.

Holt's Linear Exponential Smoothing

- Equations for Holt's Linear Exponential Smoothing

$$L_t = \alpha Y_t + (1 - \alpha)(L_{t-1} + b_{t-1})$$

$$b_t = \beta(L_t - L_{t-1}) + (1 - \beta)b_{t-1}$$

$$F_{t+k} = L_t + b_t k$$

where: L_t = estimate of the level of time series in period t
b_t = estimate of the slope of time series in period t
α = smoothing constant for level
β = smoothing constant for slope
F_{t+k} = forecast for k periods ahead
k = number of periods ahead to be forecast

Holt's Linear Exponential Smoothing

- To get the method started we need values for L_1, the estimate of the level in period 1, and b_1, the estimate of the slope in period 1.
- A commonly used approach is to set $L_1 = Y_1$ and $b_1 = Y_2 - Y_1$.

Holt's Linear Exponential Smoothing

- Example: Auger's Plumbing Service

▶ Forecast the number of plumbing jobs Auger's will have in months April through December using Holt's exponential smoothing method, with $\alpha = .1$ and $\beta = .2$.

Month	Jobs	Month	Jobs
March	353	August	409
April	387	September	399
May	342	October	412
June	374	November	408
July	396		

Holt's Linear Exponential Smoothing

- Using Smoothing Constant Values $\alpha = .1, \beta = .2$

$$L_1 = Y_1 = 353$$
$$b_1 = Y_2 - Y_1 = 387 - 353 = 34$$
$$F_2 = L_1 + b_1(1) = 353 + 34 = 387$$

$$L_2 = .1(Y_2) + .9(L_1 + b_1) = .1(387) + .9(353 + 34) = 387$$
$$b_2 = .2(L_2 - L_1) + .8(b_1) = .2(387 - 353) + .8(34) = 34$$
$$F_3 = L_2 + b_2(1) = 387 + 34 = 421$$

$$L_3 = .1(Y_3) + .9(L_2 + b_2) = .1(342) + .9(387 + 34) = 413.1$$
$$b_3 = .2(L_3 - L_2) + .8(b_2) = .2(413.1 - 387) + .8(34) = 32.42$$
$$F_4 = L_3 + b_3(1) = 413.1 + 32.42 = 445.52$$

Holt's Linear Exponential Smoothing

- Using Smoothing Constant Values $\alpha = .1, \beta = .2$

$$L_4 = .1(Y_4) + .9(L_3 + b_3) = .1(374) + .9(413.1 + 32.42) = 438.37$$

$$b_4 = .2(L_4 - L_3) + .8(b_3) = .2(438.37 - 413.1) + .8(32.42) = 30.99$$

$$F_5 = L_4 + b_4(1) = 438.37 + 30.99 = 469.36$$

$$L_5 = .1(Y_5) + .9(L_4 + b_4) = .1(396) + .9(438.37 + 30.99) = 462.02$$

$$b_5 = .2(L_5 - L_4) + .8(b_4) = .2(462.02 - 438.37) + .8(30.99) = 29.52$$

$$F_6 = L_5 + b_5(1) = 462.02 + 29.52 = 491.54$$

$$L_6 = .1(Y_6) + .9(L_5 + b_5) = .1(409) + .9(462.02 + 29.52) = 483.29$$

$$b_6 = .2(L_6 - L_5) + .8(b_5) = .2(483.29 - 462.02) + .8(29.52) = 27.87$$

$$F_7 = L_6 + b_6(1) = 483.29 + 27.87 = 511.16$$

Holt's Linear Exponential Smoothing

- Using Smoothing Constant Values $\alpha = .1, \beta = .2$

$L_7 = .1(Y_7) + .9(L_6 + b_5) = .1(399) + .9(483.29 + 29.52) = 499.95$

$b_7 = .2(L_7 - L_6) + .8(b_6) = .2(499.95 - 483.29) + .8(27.87) = 25.63$

$F_8 = L_7 + b_7(1) = 499.95 + 25.63 = 525.57$

$L_8 = .1(Y_8) + .9(L_7 + b_6) = .1(412) + .9(499.95 + 27.87) = 514.22$

$b_8 = .2(L_8 - L_7) + .8(b_7) = .2(514.22 - 499.95) + .8(25.63) = 23.36$

$F_9 = L_8 + b_8(1) = 514.22 + 23.36 = 537.57$

$L_9 = .1(Y_9) + .9(L_8 + b_7) = .1(408) + .9(514.22 + 25.63) = 524.62$

$b_9 = .2(L_9 - L_8) + .8(b_8) = .2(524.62 - 514.22) + .8(23.36) = 20.77$

$F_{10} = L_9 + b_9(1) = 524.62 + 20.77 = 545.38$

Nonlinear Trend Regression

- Sometimes time series have a curvilinear or nonlinear trend.
- A variety of nonlinear functions can be used to develop an estimate of the trend in a time series.
- One example is this quadratic trend equation:

$$T_t = b_0 + b_1 t + b_2 t^2$$

- Another example is this exponential trend equation:

$$T_t = b_0 (b_1)^t$$

Seasonality without Trend

- To the extent that seasonality exists, we need to incorporate it into our forecasting models to ensure accurate forecasts.
- We will first look at the case of a seasonal time series with no trend and then discuss how to model seasonality with trend.

Seasonality without Trend

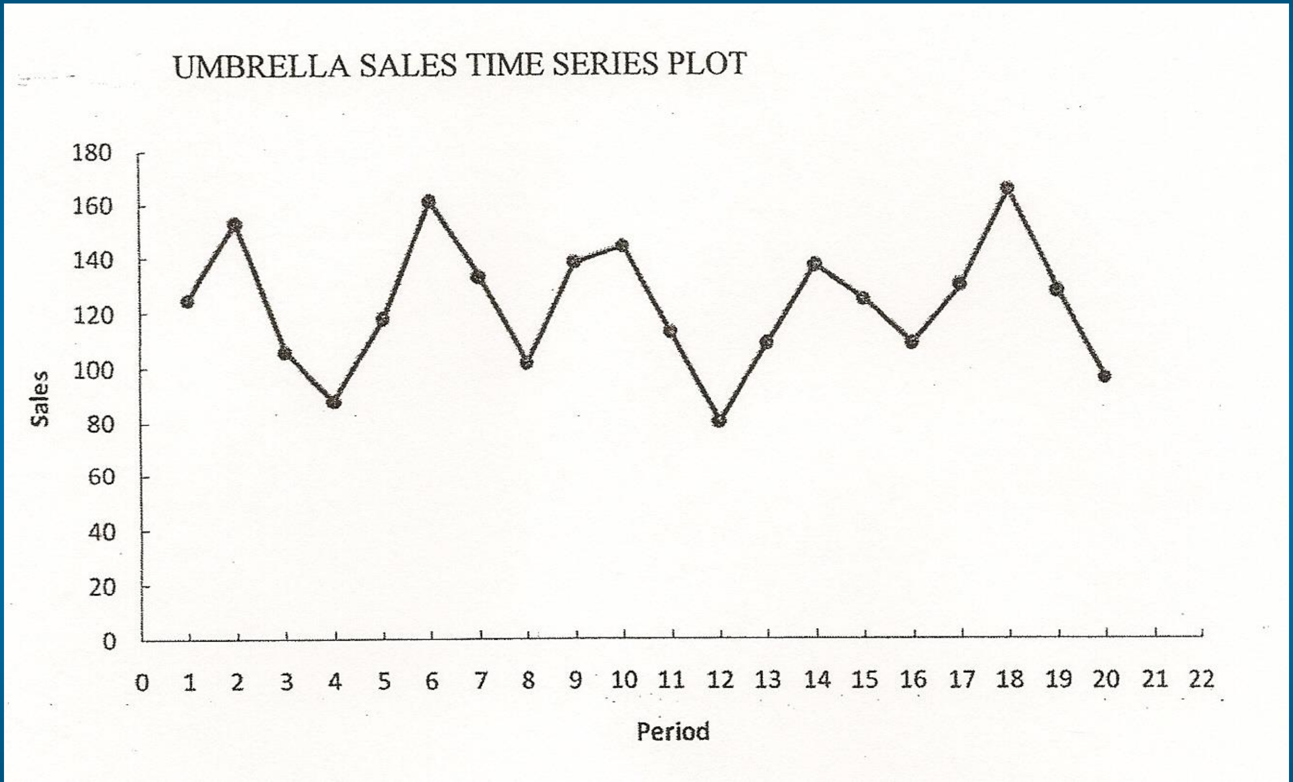
- Example: Umbrella Sales

Year	Quarter 1	Quarter 2	Quarter 3	Quarter 4
1	125	153	106	88
2	118	161	133	102
3	138	144	113	80
4	109	137	125	109
5	130	165	128	96

- Sometimes it is difficult to identify patterns in a time series presented in a table.
- Plotting the time series can be very informative.

Seasonality without Trend

- Umbrella Sales Time Series Plot



Seasonality without Trend

- The time series plot does not indicate any long-term trend in sales.
- However, close inspection of the plot does reveal a seasonal pattern.
    - The first and third quarters have moderate sales,
    - the second quarter the highest sales, and
    - the fourth quarter tends to be the lowest quarter in terms of sales.

Seasonality without Trend

- Recall from an earlier chapter that dummy variables can be used to deal with categorical independent variables in a multiple regression model.
- We will treat the season as a categorical variable.
- Recall that when a categorical variable has k levels, k – 1 dummy variables are required.
- If there are four seasons, we need three dummy variables.
  - *Qtr1* = 1 if Quarter 1, 0 otherwise
  - *Qtr2* = 1 if Quarter 2, 0 otherwise
  - *Qtr3* = 1 if Quarter 3, 0 otherwise

Seasonality without Trend

- General Form of Estimated Regression Equation is:

$$\hat{Y} = b_0 + b_1(Qtr1) + b_2(Qtr2) + b_3(Qtr3)$$

- Estimated Regression Equation is:

$$Sales = 95.0 + 29.0(Qtr1) + 57.0(Qtr2) + 26.0(Qtr3)$$

- The forecasts of quarterly sales in year 6 are:
  - Quarter 1: Sales = 95 + 29(1) + 57(0) + 26(0) = $\boxed{124}$
  - Quarter 2: Sales = 95 + 29(0) + 57(1) + 26(0) = $\boxed{152}$
  - Quarter 3: Sales = 95 + 29(0) + 57(0) + 26(1) = $\boxed{121}$
  - Quarter 4: Sales = 95 + 29(0) + 57(0) + 26(0) = $\boxed{95}$

Seasonality and Trend

- We will now extend the regression approach to include situations where the time series contains both a seasonal effect and a linear trend.
- We will introduce an additional independent variable to represent time.

Seasonality and Trend

- Example: Terry's Tie Shop
  - Business at Terry's Tie Shop can be viewed as falling into three distinct seasons: (1) Christmas (November and December); (2) Father's Day (late May to mid June); and (3) all other times.
  - Average weekly sales ($) during each of the three seasons during the past four years are shown on the next slide.

Seasonality and Trend

- Example: Terry's Tie Shop

	Season		
Year	1	2	3
1	1856	2012	985
2	1995	2168	1072
3	2241	2306	1105
4	2280	2408	1120

Determine a forecast for the average weekly sales in year 5 for each of the three seasons.

Seasonality and Trend

- There are three seasons, so we will need two dummy variables.
  - *Seas1* = 1 if Season 1, 0 otherwise
  - *Seas2* = 1 if Season 2, 0 otherwise
- General Form of Estimated Regression Equation is:

$$\hat{Y} = b_0 + b_1(Seas1) + b_2(Seas2) + b_3(t)$$

- Estimated Regression Equation is:

$$Sales = 797.0 + 1095.43(Seas1) + 1189.47(Seas2) + 36.47(t)$$

Seasonality and Trend

- The forecasts of average weekly sales in the three seasons of year 5 are:
  - Seas. 1: Sales = 797 + 1095.43(1) + 1189.47(0) + 36.47(13) = $\boxed{2366.5}$
  - Seas. 2: Sales = 797 + 1095.43(0) + 1189.47(1) + 36.47(14) = $\boxed{2497.0}$
  - Seas. 3: Sales = 797 + 1095.43(0) + 1189.47(0) + 36.47(15) = $\boxed{1344.0}$

Seasonality and Trend

- 3 estimated multiple regression equations for each quarter:

Time Series Decomposition

- Time series decomposition can be used to separate or decompose a time series into seasonal, trend, and irregular (error) components.
- While this method can be used for forecasting, its primary applicability is to get a better understanding of the time series.
- Understanding what is really going on with a time series often depends upon the use of deseasonalized data.

Time Series Decomposition

- Decomposition methods assume that the actual time series value at period *t* is a function of three components: trend, seasonal, and irregular.
- How these three components are combined to give the observed values of the time series depends upon whether we assume the relationship is best described by an <u>additive</u> or a <u>multiplicative</u> model.

Time Series Decomposition

- Additive Model
  - An additive model follows the form:

$$\boxed{Y_t = \text{Trend}_t + \text{Seasonal}_t + \text{Irregular}_t}$$

where:

Trend_t = trend value at time period t
Seasonal_t = seasonal value at time period t
Irregular_t = irregular value at time period t

  - An additive model is appropriate in situations where the seasonal fluctuations do not depend upon the level of the time series.

Time Series Decomposition

- Multiplicative Model
  - A multiplicative model follows the form:

$$\boxed{Y_t = \text{Trend}_t \times \text{Seasonal}_t \times \text{Irregular}_t}$$

where:

Trend_t = trend value at time period t

Seasonal_t = seasonal value at time period t

Irregular_t = irregular value at time period t

  - A multiplicative model is appropriate, for example, if the seasonal fluctuations grow larger as the sales volume increases because of a long-term linear trend.

Time Series Decomposition

- Example: Terry's Tie Shop

Year	Season 1	Season 2	Season 3
1	1856	2012	985
2	1995	2168	1072
3	2241	2306	1105
4	2280	2408	1120

Determine a forecast for the average weekly sales in year 5 for each of the three seasons.

Calculating the Seasonal Indexes

Step 1. Calculate the centered moving averages.

- There are three distinct seasons in each year. Hence, take a three-season moving average to eliminate seasonal and irregular factors. For example:

1st CMA = (1856 + 2012 + 985)/3 = 1617.67

2nd CMA = (2012 + 985 + 1995)/3 = 1664.00

Etc.

Calculating the Seasonal Indexes

Step 2. Center the CMAs on integer-valued periods.

- The first centered moving average computed in step 1 (1617.67) will be centered on season 2 of year 1. Note that the moving averages from step 1 center themselves on integer-valued periods because *n* is an odd number.

Calculating the Seasonal Indexes

Year	Season	Dollar Sales (Y_t)	Moving Average
1	1	1856	
	2	2012	1617.67
	3	985	1664.00
2	1	1995	1716.00
	2	2168	1745.00
	3	1072	1827.00
3	1	2241	1873.00
	2	2306	1884.00
	3	1105	1897.00
4	1	2280	1931.00
	2	2408	1936.00
	3	1120	

(1856 + 2012 + 985)/3

Calculating the Seasonal Indexes

- The centered moving average values tend to "smooth out" both the seasonal and irregular fluctuations in the time series.
- The centered moving averages represent the trend in the data and any random variation that was not removed by using the moving averages to smooth the data.

Calculating the Seasonal Indexes

Step 3. Determine the seasonal & irregular factors ($S_t I_t$).

- By dividing each actual by the moving average for the same time period, we identify the combined seasonal-irregular effect in the time series.

$$S_t I_t = Y_t / (\text{Moving Average for period } t)$$

Calculating the Seasonal Indexes

Year	Season	Dollar Sales (Y_t)	Moving Average	S_tI_t
1	1	1856		
	2	2012	1617.67	1.244
	3	985	1664.00	.592
2	1	1995	1716.00	1.163
	2	2168	1745.00	1.242
	3	1072	1827.00	.587
3	1	2241	1873.00	1.196
	2	2306	1884.00	1.224
	3	1105	1897.00	.582
4	1	2280	1931.00	1.181
	2	2408	1936.00	1.244
	3	1120		

2012/1617.67

Calculating the Seasonal Indexes

Step 4. Determine the average seasonal factors.

- Averaging all $S_t I_t$ values corresponding to that season:

Season 1:	(1.163 + 1.196 + 1.181) /3	= 1.180
Season 2:	(1.244 + 1.242 + 1.224 + 1.244) /4	= 1.238
Season 3:	(.592 + .587 + .582) /3	= .587
		3.005

Calculating the Seasonal Indexes

Step 5. Scale the seasonal factors (S_t).

- Average the seasonal factors = (1.180 + 1.238 + .587)/3 = 1.002. Then, divide each seasonal factor by the average of the seasonal factors.

Season 1: 1.180/1.002 = 1.178
Season 2: 1.238/1.002 = 1.236
Season 3: .587/1.002 = .586

3.000

Calculating the Seasonal Indexes

Year	Season	Dollar Sales (Y_t)	Moving Average	S_tI_t	Scaled S_t
1	1	1856			1.178
	2	2012	1617.67	1.244	1.236
	3	985	1664.00	.592	.586
2	1	1995	1716.00	1.163	1.178
	2	2168	1745.00	1.242	1.236
	3	1072	1827.00	.587	.586
3	1	2241	1873.00	1.196	1.178
	2	2306	1884.00	1.224	1.236
	3	1105	1897.00	.582	.586
4	1	2280	1931.00	1.181	1.178
	2	2408	1936.00	1.244	1.236
	3	1120			.586

Using the Deseasonalizing Time Series to Identify Trend

Step 6. Determine the deseasonalized data.

- Divide the data point values, Y_t, by S_t.

Using the Deseasonalizing Time Series to Identify Trend

1856/1.178

Year	Season	Dollar Sales (Y_t)	Moving Average	S_tI_t	Scaled S_t	Y_t/S_t
1	1	1856			1.178	1576
	2	2012	1617.67	1.244	1.236	1628
	3	985	1664.00	.592	.586	1681
2	1	1995	1716.00	1.163	1.178	1694
	2	2168	1745.00	1.242	1.236	1754
	3	1072	1827.00	.587	.586	1829
3	1	2241	1873.00	1.196	1.178	1902
	2	2306	1884.00	1.224	1.236	1866
	3	1105	1897.00	.582	.586	1886
4	1	2280	1931.00	1.181	1.178	1935
	2	2408	1936.00	1.244	1.236	1948
	3	1120			.586	1911

Using the Deseasonalizing Time Series to Identify Trend

Step 7. Determine a trend line of the deseasonalized data.

- Using the least squares method for t = 1, 2, ..., 12, gives:

$$T_t = b_0 + b_1 t$$

$$\text{Deseasonalized Sales}_t = 1580.11 + 33.96t$$

Using the Deseasonalizing Time Series to Identify Trend

Step 8. Determine the deseasonalized predictions.

- Substitute t = 13, 14, and 15 into the least squares equation:

$$T_{13} = 1580.11 + (33.96)(13) = 2022$$
$$T_{14} = 1580.11 + (33.96)(14) = 2056$$
$$T_{15} = 1580.11 + (33.96)(15) = 2090$$

Seasonal Adjustments

Step 9. Take into account the seasonality.

- Multiply each deseasonalized prediction by its seasonal factor to give the following forecasts for year 5:

Season 1: (1.178)(2022) = 2382
Season 2: (1.236)(2056) = 2541
Season 3: (.586)(2090) = 1225